

Individual Project - Concise Guide

What to build, document, report, and submit

ASE 420

ASE @ Northern Kentucky University

1. Goal and Scope

Build a small Python application that solves a practical problem you care about. You work independently and own the plan, implementation, tests, documentation, weekly progress, and final submission.

Success means more than working code. Your repository should show that you can define a problem, make testable requirements, plan realistic milestones, improve a design, verify quality, and communicate progress.

1.1. Start Here

1. Choose a useful problem and identify the intended user.
2. Describe the proposed solution, features, and testable requirements.
3. Create one public GitHub repository with a clear README and organized code, tests, and documentation.
4. Create two sprints with milestones and weekly goals.
5. Link the repository and documents from your individual Canvas pages.

2. Required Work

- Build a prototype, then a minimum viable product (MVP).
- Improve the design through refactoring.
- Write and run all four required test types: unit, integration, regression, and acceptance. A missing test type is graded as an incomplete deliverable and is deducted accordingly.
- You may use AI to help write and run tests, but you must verify and understand every result yourself.
- Update your Canvas progress page every week.
- Prepare a user manual and an architecture/design document in Marp Markdown and PDF.
- Publish a portfolio page with Hugo on GitHub.io.
- Complete the individual presentation or progress checkpoint scheduled on Canvas.

2.1. Weekly Update

Each weekly update should answer:

1. What milestone or goal are you working toward?
2. Are you on track?
3. What did you complete this week?
4. What will you do next week?
5. What blockers or risks exist?
6. If you are behind, what is the revised plan and recovery date?
7. Where is the supporting GitHub evidence?

3. Timeline

- **Weeks 1-3:** problem, repository, plan, requirements, architecture, test outline, and weekly schedule
- **Week 4:** finish the individual planning materials required by HW2; there is no formal individual PPP unless Canvas says otherwise
- **Weeks 5-8:** Sprint 1 - working prototype/MVP
- **Week 9:** Sprint 1 progress submission required by HW4

- **Weeks 10-15:** Sprint 2 - features, design improvement, testing, documentation, and the Canvas-scheduled presentation/checkpoint
- **Week 16:** final submission

Canvas is the official source for exact dates and presentation slots.

4. Final Deliverables

- Public GitHub repository with README.md, source code, tests, and documentation
- Project plan, requirements, milestones, and weekly schedule
- Working application
- User manual: Marp source and exported PDF
- Architecture/design document: Marp source and exported PDF
- Weekly Canvas progress reports
- GitHub.io project page and Hugo source
- Final repository ZIP without generated files, unrelated files, or hidden .git data
- ip_deliverables_checklist.md
- ip_self_eval_rubric.md

5. Grading - 150 Points

Category	Points
Plan and weekly progress	50
Requirements and features	25
Implementation	25
Tests	25
Documentation and publication	25
Total	150

Use ip_self_eval_rubric.md for the detailed criteria. The instructor may adjust the self-evaluation after reviewing the submitted artifacts and progress evidence.

6. Rules and Final Check

- Do your own work and understand everything you submit.
- AI is allowed under the course AI policy. You must be able to explain every part and independently reproduce at least 80 percent of the essential code, reasoning, or steps without AI if asked later.
- Report problems early; do not hide a missed milestone.
- Follow the deadline and extension policy on Canvas.
- Before submitting, open every link, run the application and tests, open each exported PDF, and verify the ZIP contents.